

ASSIGNMENT-3

1. Write a recursive function to check whether a string is a palindrome.
2. Write a recursive function to find the GCD of two numbers using the Euclidean algorithm.
3. Write a recursive function to count the number of digits in a number.
4. Write a recursive function to find the product of elements in a list.
5. Write a recursive function to count occurrences of a character in a string.
6. Write a recursive function to convert a decimal number to binary.
7. Write a recursive function to find the maximum element in a list.
8. Write a recursive function to generate all permutations of a string.
9. Write a recursive function to solve the Tower of Hanoi problem and print the moves.
10. Write a recursive function to generate all subsets of a set (power set).
11. Write a recursive function to solve the N-Queens problem.
12. Write a recursive function to perform binary search on a sorted list.
13. Write a recursive function to check if a number is a palindrome without converting to string.
14. Write a recursive function to flatten a nested list (e.g., [1, [2, [3, 4], 5]] → [1, 2, 3, 4, 5]).